# PGR111, solutions week 02

## Task 1 - concepts and terminology

Explain the following terminology and database concepts. (This task works best in groups.)

- DBMS
- SQL
- Relation
- Tuple
- Attribute
- Cardinality
- Domain

Write down the answers and save them. Might come in handy later in the semester? :-)

Look in the textbook and the lecture slides for answers. :-)

## Task 2 - SELECT queries

Use the MySQL example database ("schema") named **world**. (Same as we worked on last week.)

Write SQL queries to do the following:

1. List the country code, name and year of independence, sorted by year and then name, for all countries that have stated a year of independence (Hint: Something regarding NULL.)

```sql
SELECT Code, Name, IndepYear
FROM country
WHERE IndepYear IS NOT NULL
ORDER BY IndepYear, Name;
```

2. Retrieve the country code, name and continent for all countries in the Americas (both North America and South America). Sort alphabetically by country code.

```sql
SELECT Code, Name, Continent
FROM country
WHERE Continent = 'North America'
OR Continent = 'South America'
ORDER BY Code;

-- Or replace the two where-lines with this line:
-- WHERE Continent IN ('North America', 'South America')
```

3. Retrieve all language data (“columns”) for the countries that have an official language spoken by 99% or more of the population in the country. Sort by descending percentage, then alphabetically by language.

```sql
SELECT *
FROM countrylanguage
WHERE Percentage >= 99
AND IsOfficial = 'T'
ORDER BY Percentage DESC, LANGUAGE ASC;
```

4. Retrieve all country data (“columns”) for countries that have 'W' as the middle letter in their country code.

```
SELECT *
FROM country
WHERE Code LIKE '_W_';
```

5. Retrieve all columns for the countries beginning with the characters 'N', 'O' or 'P'. Sort them alphabetically by name.

```
SELECT *
FROM country
WHERE Name LIKE 'N%'
OR Name LIKE 'O%'
OR Name LIKE 'P%'
ORDER BY Name;
```

6. Sum the population of all continents, ordered by population (descending order). Show continents and population sums. (Hint: The keyword GROUP BY.)

```
SELECT Continent, SUM(Population) AS ContinentPopulation
FROM country
GROUP BY Continent
ORDER BY ContinentPopulation DESC;
```

7. Verify the answer from the previous task:
   a. Manually add together the values of task 6, above, on a calculator.
   b. Then write a new query that sums the population of all countries (no continents this time, only one value/row in the answer).

```
SELECT SUM(Population) AS WorldPopulation
FROM Country;
```

   c. Check that the numbers from a and b match.

8. Extract continents and their number of countries, for all continents.

```
SELECT Continent, COUNT(*) AS NumberOfCountries
FROM country
GROUP BY Continent;
```

9. Expand on the query in 8, above: Now extract continent and the number of countries for those continents that have 30 or more countries. Sort by number of countries in descending order. (Hint: The keyword HAVING.)

```
SELECT Continent, COUNT(*) AS NumberOfCountries
FROM country
GROUP BY Continent
HAVING NumberOfCountries >= 30
ORDER BY NumberOfCountries DESC;
```

10. Create a list of countries (their names) and their population density (number of people per square kilometer), ordered by the latter. (Hint: Look for columns containing country population and $km^2$ to get started. You can then use mathematical operators in the SELECT part of the query, to make a new column based on several table columns.)

```
SELECT Name, Population / SurfaceArea AS PopulationDensity
FROM country
ORDER BY PopulationDensity DESC;
```

## Task 3 – Like the last session, but with more/new concepts (in blue)

Use the World database and think about what data might be interesting to extract from the database. Create some queries that give you this data. The queries must only use the SQL that we have learned so far: **`SELECT`**, **`FROM`**, **`WHERE`**, **`ORDER BY`**, **`AS`**, **`COUNT`**, **`SUM`**, **`AVG`**, **`MIN`**, **`MAX`**, **`GROUP BY`**, **`HAVING`**, **`DISTINCT`**, **`LIKE`**, **`IN`**, **`BETWEEN`**, **`<WILDCARDS>`**, **`<boolean operators>`**, and only fetch from *one* table at a time. When you have succeeded in creating a query that gives the expected answer, post the task in the forum of the course. Mark the task with difficulty level "easy", "medium" or "difficult" and say something about the expected result of the query. If you cannot find the thread that will contain the week's student assignments, create it and give it a suitable name.

Once you have created some tasks and posted them in the forum, solve puzzles from others at the level of difficulty that suits you.

Feel free to solve tasks from last week as well! :-)

See Canvas for possible solutions.

## Extra

Do you have even more time? Fantastic! You can find more exercises online, or in the syllabus book. (Link to the book's website can be found in Canvas.)